Fiscal Year:	FY 2017	**Task Last Updated:**	FY 05/11/2017
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Risk of Intervertebral Disc Damage after Prolonged Space Flight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	La Jolla	**State:**	CA
Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Crew Health NNJ09ZSA002N
Start Date:	07/11/2013	**End Date:**	07/10/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Ploeger, Stephanne	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/10/2019 per NSSC information (Ed., 2/11/19)		
Key Personnel Changes/Previous PI:	May 2017: Previously listed CoInvestigator Moshe Allen is no longer working on the project.		
COI Name (Institution):	Lotz, Jeffrey Ph.D. (University of California, San Francisco) O'Neill, Conor M.D. (Self) Sayson, Jojo (Ola Grimsby Institute, San Diego) Chiang, Stephen M.D. (Methodist Hospital) Chang, Douglas M.D., Ph.D. (University Of California, San Diego) Parazynski, Scott M.D. (Arizona State University) Riascos-Castaneda, Roy M.D. (University of Texas Medical School at Houston)		
Grant/Contract No.:	NNX13AM89G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our proposal is a Flight Definition Study that will use state-of-the-art imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crewmembers before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises in space so as to establish pain and disc damage mechanisms that will serve as basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crewmember injury based on clinical findings. We hypothesize that spontaneous space-flight back pain and disc herniation are due to biomechanical and biological pathomechanisms. First, microgravity leads to higher than normal physiologic disc swelling and increased disc height that may stiffen the lumbar motion segment and cause abnormal segmental movement patterns. These biomechanical changes increase risk for annular rupture, vertebral endplate microfracture, and facet joint capsule strain. Second, increased disc swelling may alter nuclear matrix osmotic pressure and nutrient transport from endplate capillaries in adjacent vertebra. These biological changes adversely affect disc cell metabolism, causing pain and inducing disc matrix degradation. Our project directly addresses the Critical Path Roadmap Risks and Questions for NASA regarding disc injury (Integrated Research Plan (IRP) Gap-B4): Is damage to joint structure, intervertebral discs, or ligaments incurred during or following hypogravity exposure? The goal of this research is to characterize space-flight induced changes comprehensively in disc morphology, biochemistry, metabolism, and kinematics. These data will be correlated with measures of back pain intensity and disability. Crewmembers will be imaged pre-flight to establish baseline data and to characterize measurement repeatability. After long-term microgravity exposure (about 180 days on International Space Station-ISS), crewmembers will be studied while maintaining supine posture as soon as possible after return to 1-G in order to quantify the acute effects of prolonged space flight. Also, pre- and post-flight, they will be imaged supine and in standing upright posture at one body weight and again at 1.1 body weight so that MR images are obtained before and after axial body-weight loads. Pre-flight tests will be done within L-120 to L-60 days before the scheduled space flight. Post-flight tests will be undertaken 1-7 days and 30-60 days after landing to investigate re-adaptation to 1-G and to distinguish immediate and longer-term recoveries. Our proposed measures represent a comprehensive set of tests that evaluate exposure severity, potential injury mechanisms, and pain generator localization. Our research will aid understanding of spinal pain and deconditioning during prolonged microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We propose to use state-of-the-art, non-invasive imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crewmembers before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises during prolonged microgravity and after re-adaptation to Earth gravity, so as to establish pathomechanisms that will serve as a basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crewmember injury based on clinical findings. Importantly, this research will have application to back-pain patients on Earth in general and specifically, to patients exposed to long-term bed rest or lack of mobility (spinal-cord injury patients as well as patients suffering lack of exercise, mobility, and obesity). This research also has application to abnormal spinal curvature and pain suffered by children wearing heavy backpacks to and from school.
	Progress for July 2016 to June 2017 To date we have consented 12 ISS crewmembers, finished all tests on 6 ISS crewmembers and provided one-year briefings to 6 ISS crewmembers. Our IRB protocol was amended and approved by the NASA and University of California San Diego (UCSD) Human Research Committees. Highlights over the past year include (see also Bibliography section): Journal articles • Laws CJ, Berg-Johansen B, Hargens AR, Lotz JC. The effect of simulated microgravity on lumbar spine biomechanics: an in vitro study. Eur Spine J. 2016 Sep;25(9):2889-97. doi: 10.1007/s00586-015-4221-6. Epub 2015 Sep 24. PubMed PMID: 26403291. • Holt JA, Macias BR, Schneider SM, Watenpaugh DE, Lee SM, Chang DG, Hargens AR. WISE 2005: Aerobic and resistive countermeasures prevent paraspinal muscle deconditioning during 60-day bed rest in women. J Appl Physiol. 2016 May 15;120(10): 1215-22. PubMed PMID: 26893030. • Chang DG, Healey RM, Snyder AJ, Sayson JV, Macias BR, Coughlin DG, Bailey JF, Parazynski SE, Lotz JC, Hargens AR. Lumbar Spine Paraspinal Muscle and Intervertebral Disc Height Changes in Astronauts After Long-Duration Spaceflight on the International Space Station. Spine (Phila Pa 1976). 2016 Dec 15;41(24):1917-1924. PubMed PMID: 27779600. Abstract • Khieu, KK, Healey RM, Chang DG, Lotz JC, Hargens AR. Lumbar Paraspinal Muscle Atrophy and Fat Infiltration Following Prolonged Spaceflight. Proceedings of the Orthopaedic Research Society, San Diego, CA 20 March 2017. Awards • Kristine Khieu, undergraduate mentee, Second Place Award, 2016 UC San Diego Undergraduate Library Research Prize competition, co-sponsored by the UCSD Libraries, UCSD Alumni Association and the Vice-Chancellor of Student Affairs.

Task Progress:	• Lonnie Petersen, American Physiological Society Exercise & Environmental Physiology Section's National Space Biomedical Research Institute Postdoctoral Research Award, Experimental Biology, Chicago, IL, 2017. • Jeannie F. Bailey; Stephanie L. Miller; Kristine Khieu; Conor W. O'Neill; Robert M. Healey; Dezba G. Coughlin; Jojo V. Sayson; Douglas G. Chang; Alan R. Hargens; Jeffrey C. Lotz, From the International Space Station to the clinic: how prolonged unloading may elevate risk for low back pain and lumbar instability. Most Outstanding 2017 Paper in The Spine Journal. • Alan Hargens was selected for the 2017 Southwest American College of Sports Medicine Recognition Award. • Alan Hargens was selected for the Distinguished Public Service Medal, NASA's highest form of recognition that is awarded to any non-Government individual or to an individual who was not a Government employee during the period in which the service was performed, whose distinguished service, ability, or vision has personally contributed to NASA's advancement of United States' interests. The individual's achievement or contribution must demonstrate a level of excellence that has made a profound or indelible impact to NASA mission success, therefore, the contribution is so extraordinary that other forms of recognition by NASA would be inadequate. Education/Outreach • Training and mentoring of Kristine Khieu, Evan Tan, Richard Valle, and Kyle Gillespie, UCSD undergraduate students, who were finalists in NASA's Micro-g NExT competition. • UCSD Seminar Space Physiology and Exploration (WARR 87), Spaceflight induced alterations in spinal physiology presentation to UCSD freshman. • Second and successful attempt, High Altitude Balloon Launch, a NASA IVD Team Public Educational Outreach for S.T.E.M. (Science, Technology, Engineering, Math) Awareness in Kankakee County attended by Illinois 79th District Representative Kate Cloonen. • IVD co-investigators Scott Parazynski (Human Factors Design Engineer) and Jojo Sayson (Human Performance Consultant) are team members for Project Blackbird (http://www.absolutelandspeedrecord.com) to build the fastest car in the world fueled by Hydrogen Peroxide and to bring absolute land speed record back to the United States after a 32 year absence. Project Blackbird objectives on the Absolute Land Speed Record Project are: to obtain sufficient funding to build a rocket car to attempt absolute land speed record and create interest in science, technology, engineering, and mathematics by enabling US Air Force Academy Cadets, Purdue University students, and others to participate. • Jojo Sayson volunteered as resource person for a Biomedical Engineering Senior Design group at University of California, Davis tasked with developing a neck exercise machine as a countermeasure proposition intended for astronauts having neck atrophy due to the microgravity of space as well as for possible mitigation to susceptibility to neck injury (such as cervical disc herniation) after recovery on Earth. • Jojo Sayson presented an educational and empowerment talk about space travel and the solar system for 3rd grade students at Bishop MacNamara Catholic School in Bourbonnais, IL.

Bibliography Type:	Description: (Last Updated: 08/25/2025)
Abstracts for Journals and Proceedings	Khieu KK, Healey RM, Chang DG, Lotz JC, Hargens AR. "Lumbar Paraspinal Muscle Atrophy and Fat Infiltration Following Prolonged Spaceflight." 63rd Annual Meeting of the Orthopaedic Research Society, San Diego, CA, March 19-22, 2017. Proceedings of the 63rd Annual Meeting of the Orthopaedic Research Society, San Diego, CA, March 19-22, 2017. Abstract 1606. http://www.ors.org/Transactions/63/1606.pdf , Mar-2017
Articles in Peer-reviewed Journals	Laws CJ, Berg-Johansen B, Hargens AR, Lotz JC. "The effect of simulated microgravity on lumbar spine biomechanics: an in vitro study." Eur Spine J. 2016 Sep;25(9):2889-97. Epub 2015 Sep 24. PubMed PMID: 26403291; http://dx.doi.org/10.1007/s00586-015-4221-6 ; PubMed Central PMCID: PMC4808488 , Sep-2016
Articles in Peer-reviewed Journals	Holt JA, Macias BR, Schneider SM, Watenpaugh DE, Lee SM, Chang DG, Hargens AR. "WISE 2005: Aerobic and resistive countermeasures prevent paraspinal muscle deconditioning during 60-day bed rest in women." J Appl Physiol. 2016 May 15;120(10):1215-22. Epub 2016 Feb 18. http://dx.doi.org/10.1152/japplphysiol.00532.2015 ; PubMed PMID: 26893030 , May-2016
Articles in Peer-reviewed Journals	Chang DG, Healey RM, Snyder AJ, Sayson JV, Macias BR, Coughlin DG, Bailey JF, Parazynski SE, Lotz JC, Hargens AR. "Lumbar spine paraspinal muscle and intervertebral disc height changes in astronauts after long-duration spaceflight on the International Space Station." Spine (Phila Pa 1976). 2016 Dec 15;41(24):1917-24. http://dx.doi.org/10.1097/BRS.0000000000001873 ; PubMed PMID: 27779600 , Dec-2016
Awards	Khieu K. (Kristine Khieu) "Second Place Award, 2016 UC San Diego Undergraduate Library Research Prize competition, co-sponsored by the UCSD Libraries, UCSD Alumni Association and the Vice-Chancellor of Student Affairs, June 2016. " Jun-2016
Awards	Petersen L. (Lonnie Petersen) "American Physiological Society Exercise & Environmental Physiology Section's National Space Biomedical Research Institute Postdoctoral Research Award, Experimental Biology, Chicago, IL, April 2017." Apr-2017
Awards	Bailey JF, Miller SL, Khieu K, O'Neill CW, Healey RM, Coughlin DG, Sayson JV, Chang DG, Hargens AR, Lotz JC. "From the International Space Station to the clinic: how prolonged unloading may elevate risk for low back pain and lumbar instability. Most Outstanding 2017 Paper in The Spine Journal." May-2017
Awards	Hargens A. "Alan Hargens was selected for the 2017 Southwest American College of Sports Medicine Recognition Award, April 2017." Apr-2017
Awards	Hagens A. "Alan Hargens was selected for the Distinguished Public Service Medal, NASA's highest form of recognition that is awarded to any non-Government individual or to an individual who was not a Government employee during the period in which the service was performed, whose distinguished service, ability, or vision has personally contributed to NASA's advancement of United States' interests." Jan-2017